MAPPA: Symmetry-Aware Atom Mapping through Continuous Fragment Growth

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Atom mapping is often treated as if every reaction had a single correspondence between reactant and product atoms. The same endpoints can support several plausible mappings, while molecular symmetry produces many equivalent atom assignments. We introduce MAPPA, a graph-based method that preserves alternative fragment placements without explicitly listing symmetry-related bijections. Fragment growth asks only whether one compatible assignment exists; at saturation, a separate operation constructs the distinct compressed placements under the completed constraints. A cut sweep varies the growth conditions, and a separate decoder shares local bond-event responses across unique saved fragments while preserving joint symmetry constraints. A completed Golden validation recovers the annotated correspondence among the retained alternatives for 1,830 of 1,851 reactions (98.87%), with all verification outcomes resolved. Archived seed, cap and comparator studies provide additional evidence about coverage and search cost. An archived 140-reaction coordinate/Wiberg-bond-order study illustrates bond-event alternatives under unverified mappings; it is not a mapping-accuracy benchmark. MAPPA provides an inspectable representation of atom correspondence and bond changes for subsequent chemical assessment.

Date: November 1, 2026
Status: Preprint

1 Introduction

Atom-to-atom mapping assigns corresponding atoms across a reaction. It is used to identify reaction centers, extract reaction templates, and prepare endpoint structures for mechanistic calculations. A reaction may support several chemically meaningful correspondences: different mappings can imply different reac-

tion centers or bond-change patterns, while molecular symmetry can produce equivalent atom assignments. Retaining alternatives can therefore support analyses that would otherwise depend on an early choice of one mapping. The challenge is to distinguish structural alternatives from permutations that arise only from molecular symmetry.

Existing approaches use learned correspondence signals or graph-based optimization. RXNMapper derives mappings from attention in a reaction language model (15); LocalMapper learns from chemist-labeled reactions (2); and SLAPMapper combines iterative label refinement with sequential linear assignment (9). The Golden benchmark provides a common set of annotated reactions for assessing mapping recovery (11). For applications requiring alternatives, recovering the annotation somewhere in a returned family is a different objective from producing one correct top-ranked mapping.

Weighted endpoint graphs introduce a further distinction. A permissive matching tolerance can preserve a fragment despite changes in its bond orders, but the corresponding atom shuffles need not preserve the bond events assigned at a finer threshold. Replacing a compressed family by one representative can therefore hide distinct transformations. Likewise, greedily growing one fragment can prevent a neighboring fragment from reaching an alternative assignment.

MAPPA addresses these issues by branching over structural fragment choices while keeping symmetry-related atom bijections compressed (Figure 1). Growth maintains one compatible assignment; the complete compressed placements are constructed only when the fragment saturates. Default MAPPA uses one seed ordering per cut and the cut sweep. We introduce this sweep to vary the search conditions and counteract greedy fragment commitment, and also apply it to SLAP to test its benefit beyond fragment search. The final representation groups unordered matched fragment pairs while retaining their allowed mapping relations. Separate decoding collects unique fragments, computes their local bond-event responses, and composes those responses under the saved joint constraints. The method does not require the reference transformation to minimize the number of bond changes.

2 Method

Algorithm 1 connects the fragment-level operations in Figure 1. A live branch stores the fragments already placed, their atom assignments, and the compressed choices still allowed by those decisions. Growing and placing the next fragment conditions every later continuation. The matching routine is treated as an operation here; its compatibility tests and symmetry representation are defined below. For bidirectional search, the search phase is applied in each orientation, retaining the orientation of each returned family.

The forks in the search phase represent different fragment placements, not one branch for each symmetry-related bijection. The cut sweep changes the conditions of this search; a single mask can expose a mapping with several bond events. Event decoding is a separate operation on the retained output. Search caps are recorded as truncation, not treated as proof that no alternatives exist.

2.1 Weighted matching and continuous growth

Let V_R, V_P be the reactant and product atom sets, Z_R, Z_P their element labels, and W_R, W_P their symmetric bond-weight matrices. A partial mapping $m : D \subseteq V_R \to V_P$ is injective and element compatible. A mapping assigning every source atom is bijective when the endpoint element counts agree. Explicit hydrogen atoms participate in the search. Molecular coordinates can accompany the graphs but do not determine the growth decisions.

The active source graph contains edges with weight at least b. Starting from a seed atom, MAPPA grows a connected source fragment F and retains one compatible assignment as a witness. It chooses a boundary atom uniformly, then chooses one of its incident frontier edges uniformly; bond weight does not rank proposals. Let A contain the preservation edges accepted so far. A proposal adds its selected edge to A and asks whether an injective, element-compatible mapping exists satisfying

$$Z_R(u) = Z_P(m(u)), \quad W_P(m(u), m(r)) \geq b, \quad |W_R(u, r) - W_P(m(u), m(r))| \leq \tau_{\text{iso}} \tag{1}$$

Algorithm 1 MAPPA: fragment growth, compressed placements and separate decoding

Require: Oriented endpoint graphs, seed orders, optional anchors, search budgets
Ensure: Compressed final branches; optionally, local bond-event responses
$\mathcal{S} \leftarrow \varnothing$ {collected terminal branches}
for all conditions: uncut source graph and each single-edge source mask **do**
  for all requested seed orders under this condition **do**
    Initialize live branches with the supplied anchors
    repeat
      for all seeds in the current order **do**
        for all live branches eligible to grow from this seed **do**
          Grow a fragment using one-witness existence queries {Fig. 1a}
          if no admissible fragment placement is found **then**
            Retain this branch for later seeds
          else
            At saturation, construct each distinct compressed placement once
            Fork one continuation per retained placement
            Commit each placement; retain its compressed symmetry and constraints
            Enforce the cap at this hierarchy level and across live parents
          end if
        end for
      end for
    until all branches are finished or a full pass makes no progress
    Add terminal branches and their search-stop status to $\mathcal{S}$
  end for
end for
Group final branches by unordered fragment pairs; reduce repeated family records
Expose compressed branches with constraints and provenance
if bond-event decoding is requested as separate post-processing **then**
  Collect all unique fragments and their family and boundary references
  Compute shared local responses using the original matrices
  Compose requested joint choices under the saved constraints {Fig. 1c}
end if

for each $(u, r) \in A$. Only an accepted growth edge adds a preservation constraint. Other source bonds between included atoms are not implicitly required, and source nonedges do not impose negative constraints. Injectivity includes atoms assigned by established fragments.

The current witness is extended when possible. Otherwise, the existence query can change earlier assignments within the growing fragment while keeping completed-fragment assignments fixed. The same exact search either finds one witness or proves that none exists; compatibility, degree and injectivity checks eliminate impossible choices. Finding a witness accepts the edge and replaces the stored assignment. Proven absence defers the edge while other frontier edges are explored. A resource limit is not evidence of absence. Joining an established fragment preserves its locks and previously accepted constraints.

Growth continues while admissible frontier proposals remain. At saturation, completion uses all accepted constraints to construct the distinct compressed placements directly. It does not replay earlier candidate growth or enumerate symmetric bijections for later merging. Each retained placement then starts a continuation for the remaining atoms. A completed-fragment cache reuses results only when the endpoints, policies, locks, occupancy and relevant boundaries agree; eviction changes computation, not the allowed mappings.

This is a greedy search along each trajectory: an accepted extension can remove placements that would support a different fragment boundary. MAPPA broadens coverage through cut sweeps and seed orderings.

2.2 Cut sweep to vary greedy fragment boundaries

Our cut-sweep strategy reruns growth under alternative source-edge conditions. In addition to the uncut graph $G_R = (V_R, E_R)$, it searches a graph with one eligible edge e removed, $G_R^{(e)} = (V_R, E_R \setminus \{e\})$, for each

edge at or above the cut threshold. The evaluated cut threshold equals the active-graph floor $b = 0.2$, and eligible edges include bonds to H. Each subrun retains the atom set and the original endpoint weight matrices.

Removing e makes that frontier proposal unavailable, so its preservation test cannot be added through growth. An extension accepted in the uncut run may otherwise merge atoms into a fragment and exclude alternative placements. Under the cut condition, a fragment can stop at a different boundary, grow through another route, or leave atoms for a later seed. Omitting the constraint can also admit placements previously excluded by e. Thus the sweep counteracts greedy commitment by changing the conditions under which growth proceeds. It does not force the endpoints of e into different final fragments or require a bond-breaking event: they may reconnect through other source edges, and the final correspondence may preserve their bond.

MAPPA takes the union of the uncut and single-edge subruns before final branch deduplication. The default one-seed mode uses one seed ordering per subrun, so it already includes the full cut sweep. There are $|E_R|+1$ subruns at the evaluated thresholds. A single-edge cut describes the search perturbation, not the number of bond events in a returned mapping. One removed constraint can redirect subsequent fragment growth and placements throughout the reaction, exposing an alternative with several bond formations, cleavages, or bond-order changes. MAPPA therefore need not enumerate combinations of final bond events to discover multi-event alternatives. Bond-event decoding uses the original matrices, so a search cut itself contributes no event or event-count penalty. Growth remains greedy within each conditioned subrun; the union broadens coverage without making the search exhaustive.

2.3 Automorphisms and correlated mapping families

The distinction between structural branches and explicit bijections is central to the design. Even one fragment family can contain combinatorially many assignments: k independently exchangeable methyl-hydrogen triplets admit $(3!)^k$ bijections. MAPPA stores interchangeable target pools and permitted correlated actions rather than listing those bijections. The branch cap applies at every hierarchy level, including the combined live continuations across parents. It counts compressed choices, not the assignments represented inside a pool, and does not guarantee a small total search.

Completion stores a representative assignment, interchangeable target pools, fixed roles and correlated permutation actions. A pool is valid only when every represented injective assignment satisfies the completed constraints. Distinct pool allocations are grouped only when an exact automorphism of the globally colored target graph relates them under the locks and matching context. Such an action can move atoms outside the fragment together with atoms inside it; orbit labels alone do not justify independent swaps. Automorphism computation uses nauty (12).

Matching tolerance and automorphism equivalence have different roles. The relation $|x-y| \le \tau_{\text{iso}}$ is generally not transitive, so it does not by itself define a permutation group. Each accepted source edge e instead defines an exact Boolean test on a target pair (p, q),

$$\chi_e(p,q) = \mathbf{1}[W_P(p,q) \ge b \ \wedge \ |W_P(p,q) - W_R(e)| \le \tau_{\text{iso}}]\,. \tag{2}$$

Interchangeability requires the same eligibility and support for these tests, together with the matching context. Completion depends on the finished constraints, rather than on how temporary witnesses changed during growth. Local pools and global correlated actions serve different purposes; neither can remove a required constraint.

The saved mapping relation also contains the constraints imposed by its search decisions. We represent a flat family $\mathcal{F}_j$ by a seed mapping m_j, a sequence of allowed action groups $G_{j,1}, \ldots, G_{j,k}$, and final preservation constraints C_j:

$$\mathcal{F}_j = \{a_k \circ \cdots \circ a_1 \circ m_j : a_i \in G_{j,i},\ C_j(a_k \circ \cdots \circ a_1 \circ m_j)\}\,. \tag{3}$$

These actions encode correlated moves; atoms in a common orbit are not independently permutable. Disjoint actions commute and can be placed in a canonical order. The order of overlapping actions is retained. A sequence of groups is not generally equivalent to the group generated by their union.

The action sequence includes the allowed pool choices and correlated moves, conditioned on prior locks. Different constraints or action programs can define different families even when their representative atom

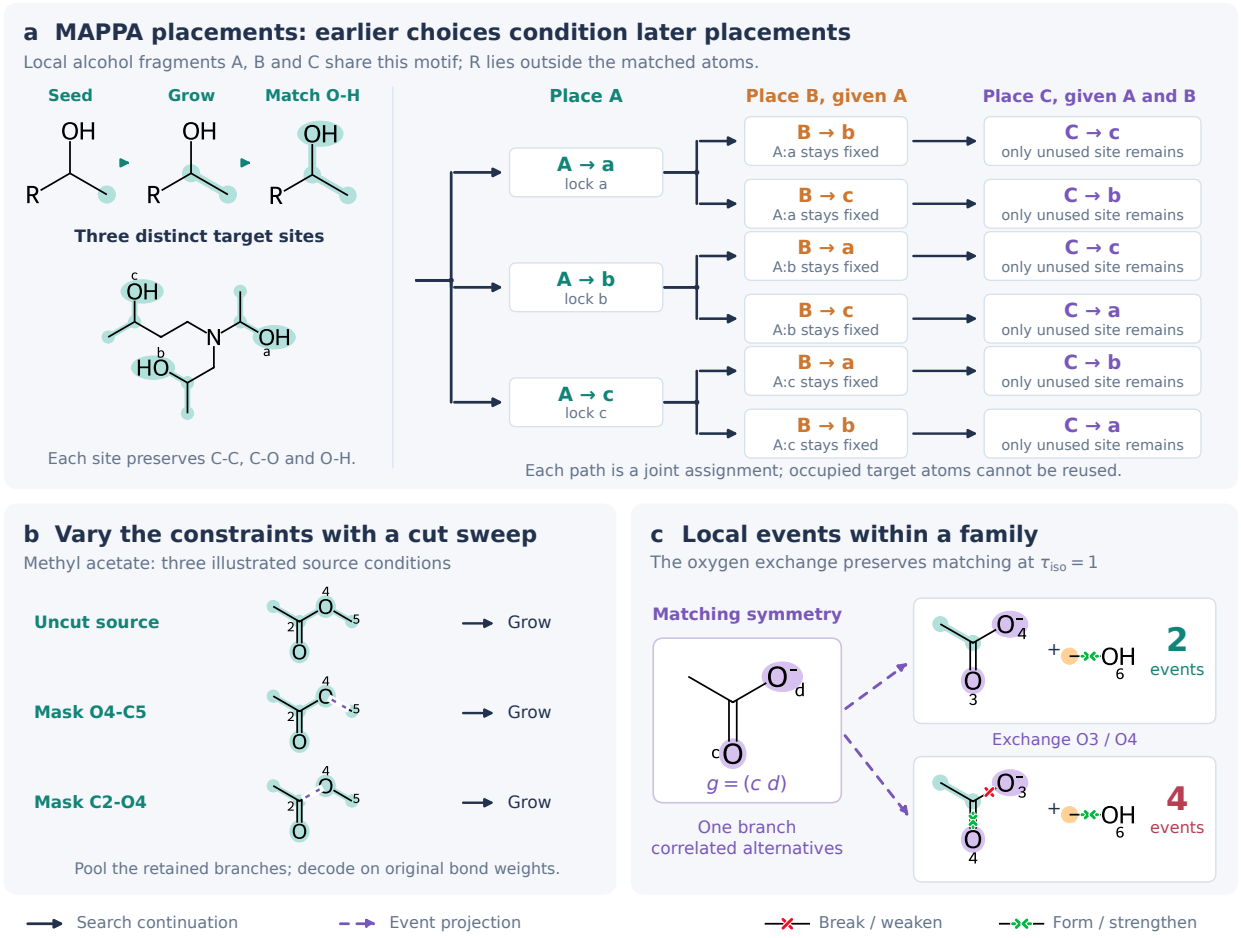


Figure 1 MAPPA: conditional fragment branching and compressed symmetry. (a) A local alcohol motif grows while retaining its O–H bond. Three sites in the depicted target have distinct molecular environments. For three source fragments sharing this motif, placing A fixes one site; B then branches over the two unused sites, and C takes the remaining site. The tree illustrates local conditioning and injectivity; R denotes atoms outside the matched subgraph, and this is not a complete reaction-search trajectory. All attached hydrogens in the illustrated local matches are checked. (b) The cut sweep searches the uncut source graph and graphs with one source edge masked. The illustrated ester masks change the growth and preservation constraints; their outputs are pooled. Masking an edge does not itself assign a bond event. (c) Exchanging the two acetate oxygen images preserves the fragment's element and bond-weight matching tests at $\tau_{\text{iso}} = 1$, but changes two heavy-atom events into four at threshold 0.5. Neither acetate oxygen carries H. Solid black lines retain the endpoint bonds. Red × marks on the bonds indicate cleavage or weakening; paired green arrows pointing inward indicate formation or strengthening. Pastel colors track fragment membership or symmetry sites. Dashed arrows are event projections within a family, not additional search branches. Saved constraints govern allowed symmetry actions. These formal-bond-order event signatures illustrate MAPPA decoding, not distinct mechanisms or resonance-normalized transformations.

sets agree. MAPPA retains these distinctions; sharing a fragment description does not enlarge its saved mapping relation.

2.4 Unordered final branches

During search, history and boundary conditions matter because they determine valid continuation. At completion, the branch identity is instead the unordered collection of matched atom sets,

$$\mathcal{B} = \{(R_1, P_1), \ldots, (R_q, P_q)\}, \qquad P_i = m(R_i), \tag{4}$$

under a fixed matching policy. Sorting atoms within each set and then sorting the fragment pairs removes dependence on growth order and fragment labels. Growing A before B or B before A yields one branch when the final paired sets agree. The colors within each candidate in Figure 2 illustrate these matched fragment pairs.

Each branch stores the union of its flat families. Identical normalized family records are shared, while distinct constraints or action programs remain available. This transformation preserves the represented mapping union: it changes its grouping, not its admissible assignments. Provenance links retain the discovering paths, but the original search hierarchy is unnecessary for decoding. This conservative deduplication does not assert that the remaining families are mutually disjoint.

2.5 Decoding from unique fragments

For an unordered source pair $\{u, v\}$ whose atoms are both mapped, define the signed change by

$$\Delta_{uv}(m) = W_P(m(u), m(v)) - W_R(u, v), \qquad s_{uv}(m) = \begin{cases} +1, & \Delta_{uv}(m) > \delta_{uv}, \\ -1, & \Delta_{uv}(m) < -\delta_{uv}, \\ 0, & \text{otherwise.} \end{cases} \tag{5}$$

Positive events denote bond formation or strengthening; negative events denote cleavage or weakening. A change exactly equal to the threshold is not an event. The thresholds are 0.3 for pairs involving a metal and 0.5 otherwise, independently of τ_{iso}. Hydrogen atoms participate. For partial mappings, bonds crossing from mapped to unmatched atoms contribute losses on the source side and gains on the target side, without inventing an assignment for the unmatched atoms. Bonds wholly inside unmatched material are omitted. The event count $E(m)$ sums the nonzero responses, counting each bond once. All responses use the original endpoint matrices, including bonds omitted during the cut sweep.

Decoding starts after family reduction across all supplied seeds and cuts for one reaction and direction. It first collects the complete inventory of unique saved fragments, their occurrences in families, and their constraints, actions and boundary references. It then computes shared responses for internal bonds, bonds between fragments, and mapped/unmatched boundaries. Two occurrences share a response only when the relevant fragment, witness, ordered actions and boundary context agree. A cut is a search condition, not an independent decoding unit.

For a requested joint choice, the decoder validates the saved family constraints and combines the corresponding local responses. A shared symmetry action remains one correlated choice across every affected fragment. Alternative fragment partitions describe the same family and are not added together. This separates reusable local calculations from branch composition without rematching or expanding complete bijections. Matching-preserving shuffles can still yield different events, as in Figure 1c.

Ordinary decoding retains this local representation; it does not automatically enumerate distinct whole-reaction event classes or select a minimum-event mapping. Those are separate queries on the retained families. The archived coordinate evaluation used an explicit event-window query: event-invariant families were certified directly, and remaining choices were encoded in Z3 (5) to obtain one witness per signed event class. Exhaustion or exclusion of every retained family certified completeness within $E \leq K$; timeouts remained unresolved. That certificate concerns saved search output within a window, not exhaustive structural search or the completeness of ordinary local decoding.

2.6 Alternative correspondences and chemical interpretation

An atom mapping summarizes the net correspondence resulting from a chemical process without specifying its intermediates or elementary steps. When that pathway is unknown, endpoint structures alone may not determine a unique correspondence. MAPPA retains mapping families and their correlated choices so that their bond events and conditional atom assignments can be queried. An explicit event-class query can supply representative witnesses and supporting families. Distinct patterns can have equal event counts, and minimizing the count need not recover the reference correspondence.

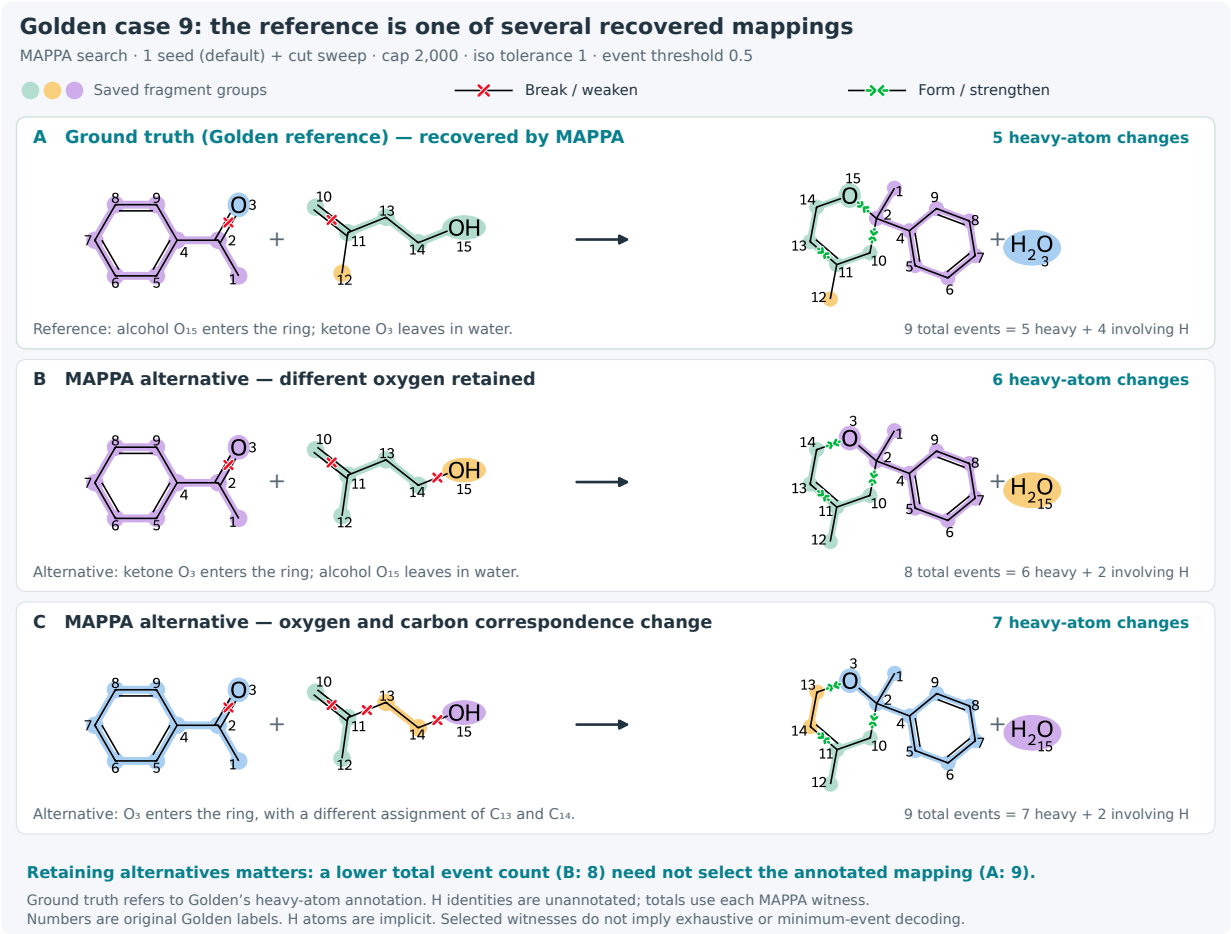


Figure 2 A recovered reference and alternative correspondences from Golden case 9. A matches all 15 annotated heavy-atom pairs. B exchanges the oxygen origins; C also exchanges C_{13}/C_{14}. Pastel colors identify matched fragments. Black lines show endpoint bonds; red × marks indicate cleavage or weakening, and green inward arrows indicate formation or strengthening. H atoms are implicit in the drawing; event totals include H. Each witness minimizes H-involving changes for its fixed heavy-atom correspondence. "Ground truth" denotes the dataset annotation, not a verified pathway. The three archived witnesses come from one-seed MAPPA sweep search at cap 2,000 and $\tau_{\text{iso}} = 1$, without reference labels; events use threshold 0.5.

Figure 2 illustrates both points with saved Golden case 9 witnesses. The annotated correspondence places the alcohol oxygen in the product ring and leaves the ketone oxygen in water (A). Another candidate exchanges these oxygen origins (B); a third also exchanges two carbon assignments (C). Candidate B has fewer total events than the reference, while A and C have equal totals but different correspondences. Thus, selecting either one bijection or only the lowest event count discards information about the alternatives. These are selected witnesses from the archived search, not an exhaustive decoded set or validated reaction pathways.

3 Evaluation

3.1 Datasets and comparison protocol

We report a completed validation of the one-witness/direct-completion pipeline on all 1,851 Golden reactions (11), alongside archived September studies of seed count, branch caps, comparator performance and 140 coordinate/WBO endpoint pairs. The archived studies used their recorded implementation and are not reruns of the current method. Their measurements and witnesses are retained with that scope.

Table 1 Archived evaluation configurations. Both use $b = 0.2$, $\tau_{\text{iso}} = 1$, explicit H, and uncut plus single-edge searches. One seed per cut and the sweep define the default setting. Archived Golden measurements cover sweep search; no-sweep controls retain only the uncut graph. A seed setting counts source-atom orderings per cut.

Setting	Golden	Coordinate/WBO
Reactions	1,851	140
Search directions	Bidirectional	Reactant to product
Seed orderings	1 (default), 2, 3, 10	1 (default)
Branch cap	100 (default)	100 (default), 2,000
Evaluation	Reference-family recovery	Signed event-class coverage

Golden inputs use formal bond orders and explicit H, with reference labels removed before search. Recovery requires the annotated heavy-atom correspondence, including unmatched atoms, modulo endpoint chemical symmetries preserving charge, isotope, H count, and stereochemistry. These chemical colors define reference equivalence; they do not imply that element/WBO matching itself enforces every chemical attribute. Failures and unresolved cases remain in the denominator. This measures reference inclusion among returned alternatives, not their ranking.

LocalMapper provides the prior accuracy-SOTA baseline (2). Its published Golden accuracy was 89.8% on 1,758 retained reactions with manual calibration. SLAP reported 86.9% on that dataset version, counting reference inclusion among its candidates (9). These literature scores are separate from our strict re-evaluation. We additionally apply the cut sweep introduced here to SLAP; "SLAP sweep" denotes this extension, not a published SLAP score. It is our strongest evaluated prior-method baseline for alternative recovery. Released RXNMapper, Chython, Indigo, and RDT provide further comparisons (15; 3; 6; 14).

The coordinate collection contains 18–137 explicit atoms per endpoint, including organometallic and multi-component structures. It was inspected during development and has no verified atom mappings: it is not a mapping-accuracy benchmark. The archived study compared the lowest event count found and the patterns achieving that count. MAPPA used supplied WBO matrices; SLAP used XYZ-derived binary graphs. Returned SLAP heavy-atom assignments received hydrogen refinement and were scored on the same WBO matrices as MAPPA. The comparison covers those saved witnesses, not all possible SLAP hydrogen assignments or a fresh current-implementation evaluation.

3.2 Completed validation of the current matching and decoding design

The completed Golden run processed 1,851 reactions and 3,702 directions through matching, family reduction, local decoding and separate reference verification. With one seed ordering, random seed 42, cap 100, full bidirectional cut sweeps, $b = 0.2$ and $\tau_{\text{iso}} = 1$, it recovered 1,830 references (98.87%) and 3,618 directional references. All tasks completed, and every verification outcome was resolved. Recovery means membership in the retained mapping relation; a nonrecovery does not prove that no compatible mapping exists beyond the bounded search.

This run validates the one-witness/direct-completion matcher and unique-fragment decoder together at its recorded source version. Its per-case outcomes and stage timings are retained separately from the archived studies below. The reported recovery is not a claim that every later revision has undergone a new full benchmark.

3.3 Archived evaluation configurations

Table 1 lists the archived settings. Search did not use reference or comparator correspondences. Full configurations, implementation versions, and reproduction scripts are provided in the repository (Appendix A).

Table 2 Golden reference recovery and runtime in the archived evaluation. Recovery uses all 1,851 primary-evaluation records. MAPPA retains compressed families; SLAP returns explicit candidates; other methods return one bijection over their matched atoms. MAPPA uses cap 100 and bidirectional search. Bidirectional SLAP combines both directions and both binary/weighted modes; the default SLAP row uses binary only. Times are mean CPU seconds per reaction, excluding separate bond-event decoding. Unmarked times use the same 1,403 completed reactions on an AMD EPYC 9J14. *a*: archived completed calls (1,718–1,851 per method), CPU model unrecorded; these times are not directly comparable with unmarked times. †: LocalMapper, prior accuracy SOTA (2024). ‡: SLAP with our sweep, strongest evaluated prior-method recovery baseline.

Method	Search setting	Output	Reference recovered	Mean CPU s
MAPPA	1 seed, sweep (default)	Families	1,834 (99.08%)	0.963
MAPPA	2 seeds, sweep	Families	1,834 (99.08%)	1.619
MAPPA	3 seeds, sweep	Families	1,837 (99.24%)	2.265
MAPPA	10 seeds, sweep	Families	1,840 (99.41%)	6.993
MAPPA	1 seed, no sweep	Families	1,489 (80.44%)	0.040
SLAP, bidirectional	No sweep	Candidates	1,661 (89.74%)	0.105
SLAP, bidirectional‡	Sweep	Candidates	1,796 (97.03%)	2.793
SLAP, binary	Default, no sweep	Candidates	1,591 (85.95%)	0.075[a]
RXNMapper	Default, no sweep	One bijection	1,547 (83.58%)	0.056[a]
LocalMapper†	Default, no sweep	One bijection	1,605 (86.71%)	0.294[a]
Chython	Default, no sweep	One bijection	1,561 (84.33%)	0.612[a]
Indigo	Default, no sweep	One bijection	692 (37.39%)	0.184[a]
RDT	Default, no sweep	One bijection	1,030 (55.65%)	1.021[a]

3.4 Archived Golden reference recovery and search cost

In the archived campaign, the one-seed MAPPA configuration recovered 99.08% of Golden references, compared with 97.03% for SLAP sweep (Table 2 and figure 3). These scores belong to that campaign, separate from Section 3.2. The paired comparison includes 43 references recovered only by MAPPA and 5 only by SLAP. LocalMapper is the strongest evaluated method returning one bijection. Additional MAPPA seeds yielded modest recovery gains at increasing search cost; two seeds did not improve recovery over one.

The paired archived controls attribute 345 additional reference recoveries to the MAPPA sweep and 135 to the SLAP sweep. Without the sweep, one-seed MAPPA had lower coverage than bidirectional SLAP. This supports varying the growth constraints to expose alternatives excluded by greedy fragment extension (Section 2.2); symmetry expansion alone cannot restore discarded placements. These differences are not measurements of a current no-sweep control.

Smaller-first grows fragments from the reaction side with fewer explicit atoms (including H) and matches into the other side; larger-first reverses these roles. In the archived comparison, combining both directions added 31 recoveries over smaller-first alone, demonstrating complementary alternatives (Table 3).

In these archived measurements, one-seed MAPPA had a lower mean and median search cost than SLAP sweep, but a higher 95th percentile. Two seeds approximately doubled the mean cost without improving recovery. These Golden timings cover search, excluding reference verification and separate bond-event decoding. The two timing tables use different experiments and should be compared within their stated CPU and reaction sets; neither establishes a current-implementation speed ranking.

In the archived cap study, increasing the branch cap from 100 to 2,000 yielded no additional swept recoveries among cases resolved at both caps, while increasing cost and unresolved outcomes at higher seed counts (Table A3). The uncut search gained 19 confirmed recoveries. Thus, in that study a larger branch budget did not substitute for changing the growth conditions.

The default-comparator rows in Table 2 use released interfaces without prediction repair or reference-guided reranking. Binary SLAP is retained as the stronger default configuration. The other methods return at most

Table 3 Archived recovery and runtime by search orientation. All rows use the sweep. Recovery uses all 1,851 Golden records; CPU statistics are seconds per reaction over the same 1,821 completed reactions (Apple M4 Pro CPU). Smaller-first grows from the side with fewer explicit atoms, including H; larger-first reverses the source and target. Equal-size ties use stored order and its reverse. Bidirectional output combines both searches, and its CPU cost is their sum. SLAP combines binary and weighted modes. CPU/SLAP compares means within each orientation. Timing scopes are specified in Appendix A.

Method	Orientation	Recovered	Mean	Median	95th pct.	CPU/SLAP
MAPPA, 1 seed (default)	Smaller-first	1,803 (97.41%)	1.141	0.263	5.289	0.34
	Larger-first	1,779 (96.11%)	1.377	0.348	6.401	0.92
	Bidirectional	1,834 (99.08%)	2.518	0.719	12.913	0.52
MAPPA, 2 seeds	Smaller-first	1,815 (98.06%)	2.232	0.416	10.635	0.67
	Larger-first	1,805 (97.51%)	2.721	0.578	14.064	1.83
	Bidirectional	1,834 (99.08%)	4.953	1.250	26.350	1.03
SLAP sweep (prior baseline)	Smaller-first	1,775 (95.89%)	3.329	0.422	4.752	1.00
	Larger-first	1,770 (95.62%)	1.490	0.455	4.568	1.00
	Bidirectional	1,796 (97.03%)	4.819	0.881	11.048	1.00

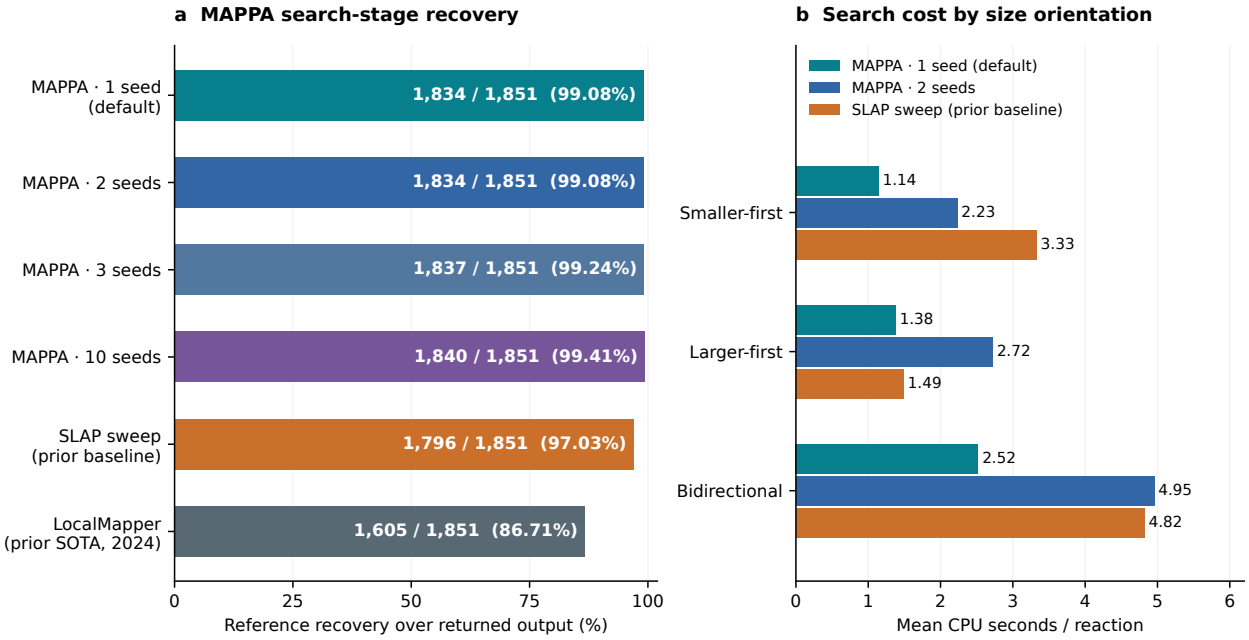


Figure 3 Archived Golden reference recovery and search cost. (a) Reference recovery over all 1,851 reactions. MAPPA and SLAP use our cut sweep; LocalMapper is the prior accuracy-SOTA baseline and returns one bijection. All scores use the archived evaluator. (b) Mean search CPU over the same 1,821 completed reactions, by orientation. Bond-event decoding is excluded. Full comparisons and timing scopes are in Tables 2 and 3.

one explicit bijection per reaction, whereas SLAP returns multiple candidates. Invalid endpoint chemistry and duplicate atom labels count as failures; the repository records their frequencies. These results describe the evaluated interfaces and scoring rule, including its difference from RDT's chemical-change criterion.

3.5 Archived bond-event comparison on 140 coordinate reactions

The archived coordinate study used one-seed MAPPA sweep search with a branch cap of 2,000 and returned complete mappings for all 140 reactions. At cap 100, cases 123 and 125 returned no complete mapping.

The 2,000-branch configuration therefore supplies this comparison, while 100 remains the library default. These completion counts and the event-window results below have not been remeasured with the current implementation.

The archived MAPPA output matched or lowered the minimum event count found by both native SLAP and SLAP sweep in every reaction (Table 4 and figure 4). This does not imply recovery of every alternative: at equal minima, six SLAP-sweep patterns remained absent across four reactions. An equal count can describe different changed bonds, and a lower count alone does not establish a more plausible transformation.

Table 4 Archived minimum bond-event comparison on 140 reactions. MAPPA uses one-seed sweep search at cap 2,000. The first three columns count reactions by MAPPA's minimum event count relative to the comparator. The final column gives comparator minimum-event patterns recovered by MAPPA among reactions with equal minima. Counts refer to returned candidates, not proven global optima.

Compared with	MAPPA fewer	Equal	MAPPA more	Patterns at equal minima
SLAP sweep	4	136	0	158/164
Native SLAP	15	125	0	138/140

To compare alternatives above MAPPA's own minima, the archived study queried through one event above the higher of the two SLAP minima for each reaction. MAPPA retained 162 of 168 SLAP-sweep minimum-event patterns and 151 of 160 native-SLAP patterns within these windows. The output contains 300 distinct patterns, including 166 at MAPPA's own minima. Every saved family was exhausted or certified within its window; this establishes completeness for those retained families, not exhaustive structural search.

Table 5 Archived stage CPU time per reaction on the 140 coordinate reactions. Measurements use an Apple M4 Pro CPU. MAPPA uses one-seed sweep search at cap 2,000. These stage costs exclude MAPPA decoding and do not establish an end-to-end speed ranking. SLAP refinement covers its returned witnesses. Timing boundaries are specified in Appendix A.

Recorded stage	Mean CPU s	Median CPU s	95th pct. CPU s
MAPPA sweep search	1.047	0.314	3.481
SLAP sweep mapping	1.337	0.546	5.919
SLAP H refinement / initial scoring	0.053	0.011	0.207

The stage timings in Table 5 cover different outputs: MAPPA search retains families for subsequent decoding, while SLAP refinement operates on returned witnesses. They therefore do not establish an end-to-end speed ranking.

The archived September implementation completed search and event-window decoding for 140/140 coordinate reactions. On this full set, the full pipeline takes a mean 6.07 CPU seconds per reaction (median 0.81; 95th percentile 25.47). Decoding accounts for a mean 4.65 CPU seconds (median 0.29). These measurements include search, final-family construction, decoding, and candidate output; they are separate from the search-only comparison above (Table A4).

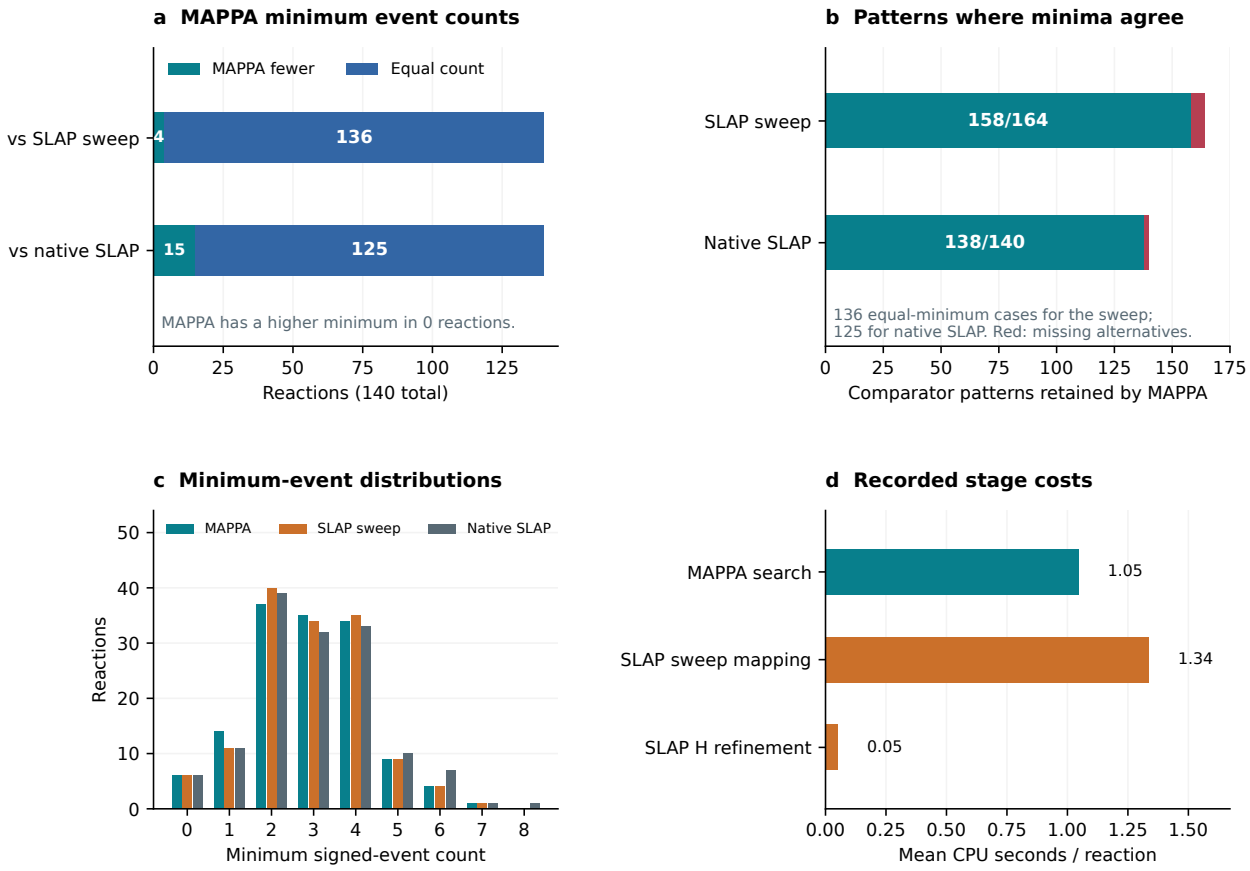


Figure 4 Archived bond-event comparison on 140 coordinate reactions. MAPPA uses one seed, the sweep, and cap 2,000. (a) Reactions with fewer or equal MAPPA events than each SLAP baseline. (b) Comparator minimum-event patterns recovered by MAPPA at equal minima. (c) Minimum-event count distributions. (d) Mean CPU for search and SLAP refinement, excluding MAPPA decoding (Table 5). Atom mappings are unverified; this compares bond events, not mapping accuracy.

4 Discussion

4.1 Applications and future directions

Atom correspondence is a preprocessing decision in reaction-path interpolation: the assignment determines which product position each reactant atom approaches. Atomic-index assignment is a recognized obstacle in reaction-pathway search (4). MAPPA's complete correspondences and allowed joint shuffles could be used to reorder and align endpoints before constructing initial paths for nudged elastic band (NEB) calculations (8). Different event patterns provide alternative reaction cores, while symmetry-related assignments within one pattern can give different interpolations. This offers a systematic way to prepare several TS-search starting points. Mapping the same core into a TS guess also allows its vibrational modes to be assessed against the intended bond changes.

Structure verification is another application. A generative model may return a molecule with reordered atoms or a different conformation, making direct coordinate comparison unsuitable for checking its connectivity. AAM supplies an explicit correspondence under which the target and generated structures can be compared for missing or extra connections. A complete witness that passes this comparison certifies agreement under the chosen graph representation and tolerances. Connectivity alone does not establish stereochemical identity or energetic stability, and failure to find a witness in a bounded search is not proof of a mismatch.

Such checks could also supply a verification mechanism for large language model (LLM) post-training, including reinforcement learning with verifiable rewards (RLVR) (10). For tasks that ask a model to generate a target structure or satisfy specified structural constraints, an independently checked atom correspondence could provide a task-specific reward. The reward would certify the stated structural conditions, with unresolved searches kept distinct from verified failures. This is a proposed use of the verifier; we have not evaluated an RLVR training loop or its effect on molecular generation.

For pathway and retrosynthesis analysis, alternative event patterns expose different atom origins, reaction centers, and possible fragment disconnections. These could help organize mechanistic hypotheses or compare proposed precursors while retaining symmetry-related choices for subsequent geometric analysis. One net correspondence can admit several intermediate sequences, however, and a compatible mapping need not describe a feasible reaction. Energies, reaction conditions, stereochemistry, and experimental evidence remain necessary to assess the proposed routes. The present evaluation concerns correspondence coverage and bond-event alternatives; it does not measure improvements in NEB convergence, TS-search success, or synthesis planning.

Alternative enumeration is not unique to MAPPA or SLAP; for example, an Ising-computing approach enumerates optimal mappings and reduces them by molecular symmetry (1). MAPPA's contribution is to retain conditional fragment families and expose their distinct event patterns and allowed shuffles without expanding all bijections.

4.2 Coverage and remaining search limitations

An audit of all eleven ten-order Golden misses from the archived study separates unmatched-atom decisions from fragment-placement failures (Table A2). It is not an audit of the separate current one-seed validation. In four archived reactions, every retained family maps exactly one more heavy-atom pair than the reference. Such a family cannot reproduce the reference's unmatched-atom pattern through symmetry shuffling: permutations preserve the number of assigned pairs. The remaining seven lack the required correspondence within their saved families. Re-ranking or expanding those families into explicit bijections cannot supply a missing structural alternative.

Two archived traces show how greedy extension excluded a representable reference. One reference exchanges adjacent labels in an otherwise unchanged pyridine ring; another exchanges the carbonyl-oxygen attachments of oxalic acid across its retained carbons. Growth preserved bonds that these correspondences require separating. Uncapped trajectories still missed both references, while reference-constrained fragment choices established their representability. In the oxalic-acid example, a reference-selected three-edge relaxation also exposed the reference to matching; the single-edge sweep did not contain that cut set. These are feasibility diagnostics, not additional benchmark recoveries or traces of the current matcher. For the other five correspondence misses, the audit did not isolate cap truncation from unsampled growth choices. They motivate selective alternatives to successful growth extensions and explicit choices about leaving atoms unmatched, rather than treating larger caps or more symmetry expansion as sufficient remedies.

Only one of the eleven misses is recovered by any tested comparator configuration, including the SLAP sweep. The shared failures warrant inspection of the supplied correspondences as well as search improvements; they do not establish annotation errors. All eleven remain in the reported denominator. Independent evaluation on annotated coordinate reactions is needed to assess generalization beyond the development collection.

Code and data

Code and reproduction instructions are available at MAPPA source repository, branch `main`. The repository guide `manuscript/REPRODUCIBILITY.md` links full benchmark configurations, software versions, per-case results, and figure-generation scripts. Golden is distributed by its original authors (11).

Acknowledgements

We gratefully acknowledge the longstanding contributions of the Matter Lab's current and past group members (matter.toronto.edu), and in particular those of the El Agente team (elagente.ca). C.C. acknowledges support from the DOE grant with the University of Minnesota, award #A006801504, and the Basic Science Research Program through the National Research Foundation of Korea (NRF), funded by the Ministry of Education (RS-2025-02634334). Y.K. was supported by the CIFAR AI Safety Catalyst Award (Catalyst Fund Project #CF26-AI-001). J.B.P.S. acknowledges funding of this project by the Natural Sciences and Engineering Research Council of Canada (NSERC) Alliance Grant #ALLRP587593-23 (Quantamole). A.W. acknowledges support from the Canada Graduate Scholarships—Doctoral (CGS-D) program.
This research is part of the University of Toronto's Acceleration Consortium, which receives funding from the CFREF-2022-00042 Canada First Research Excellence Fund. Computations were performed on the Niagara supercomputer at the SciNet HPC Consortium. SciNet is funded by the Canada Foundation for Innovation; the Government of Ontario; the Ontario Research Fund—Research Excellence; and the University of Toronto. A.A.-G. thanks Anders G. Frøseth for his generous support and acknowledges the generous support of NSERC and the Canada 150 Research Chairs program. A.A.-G. and V.B. also acknowledge the generous support of Natural Resources Canada and the Canada 150 Research Chairs program and the University of Toronto's Acceleration Consortium, which receives funding from the CFREF-2022-00042 Canada First Research Excellence Fund.
This work was supported by the Defense Advanced Research Projects Agency (DARPA) under Agreement No. HR0011262E022. This work was also supported by the AI2050 program of Schmidt Sciences.
Molecular graphics and analyses were performed with UCSF ChimeraX (7; 13), developed by the Resource for Biocomputing, Visualization, and Informatics at the University of California, San Francisco, with support from National Institutes of Health R01-GM129325 and the Office of Cyber Infrastructure and Computational Biology, National Institute of Allergy and Infectious Diseases.
We thank Kjell Jorner and Robert Pollice for helpful discussions.

References

[1] Mohammad Ali, Yuta Mizuno, Seiji Akiyama, Yuuya Nagata, and Tamiki Komatsuzaki. Enumeration approach to atom-to-atom mapping accelerated by Ising computing. *Journal of Chemical Information and Modeling*, 65 (4):1901–1910, 2025. doi: 10.1021/acs.jcim.4c01871.

[2] Shuan Chen, Sunggi An, Ramil Babazade, and Yousung Jung. Precise atom-to-atom mapping for organic reactions via human-in-the-loop machine learning. *Nature Communications*, 15:2250, 2024. doi: 10.1038/s41467-024-46364-y.

[3] Chython contributors. Chython: molecular and reaction processing toolkit. https://github.com/chython/chython. Evaluated version 2.18 with chython-rxnmap 2.0; model hash archived.

[4] Deb Sankar De, Marco Krummenacher, Bastian Schaefer, and Stefan Goedecker. Finding reaction pathways with optimal atomic index mappings. *Physical Review Letters*, 123:206102, 2019. doi: 10.1103/PhysRevLett.123.206102.

[5] Leonardo de Moura and Nikolaj Bjørner. Z3: An efficient SMT solver. In *Tools and Algorithms for the Construction and Analysis of Systems*, pages 337–340. Springer, 2008. doi: 10.1007/978-3-540-78800-3_24.

[6] EPAM Systems. Indigo cheminformatics toolkit. https://github.com/epam/Indigo. Evaluated version 1.46.0.

[7] Thomas D. Goddard, Conrad C. Huang, Elaine C. Meng, Eric F. Pettersen, Gregory S. Couch, John H. Morris, and Thomas E. Ferrin. UCSF ChimeraX: Meeting modern challenges in visualization and analysis. *Protein Science*, 27:14–25, 2018. doi: 10.1002/pro.3235.

[8] Graeme Henkelman, Blas P. Uberuaga, and Hannes Jónsson. A climbing image nudged elastic band method for finding saddle points and minimum energy paths. *The Journal of Chemical Physics*, 113(22):9901–9904, 2000. doi: 10.1063/1.1329672.

[9] Shin-ichi Koda and Shinji Saito. General and scalable atom-to-atom mapping via Weisfeiler–Lehman-like approximate graph matching. *ChemRxiv*, 2025. doi: 10.26434/chemrxiv-2025-hthwn. Preprint, version 1, 25 November 2025.

[10] Nathan Lambert, Jacob Morrison, Valentina Pyatkin, et al. Tulu 3: Pushing frontiers in open language model post-training. arXiv:2411.15124, 2024.

[11] Arkadii Lin, Natalia Dyubankova, Timur I. Madzhidov, Ramil I. Nugmanov, Jonas Verhoeven, Timur R. Gimadiev, Valentina A. Afonina, Zarina Ibragimova, Assima Rakhimbekova, Pavel Sidorov, Andrei Gedich, Rail Suleymanov, Ravil Mukhametgaleev, Joerg Wegner, Hugo Ceulemans, and Alexandre Varnek. Atom-to-atom mapping: A benchmarking study of popular mapping algorithms and consensus strategies. *Molecular Informatics*, 41(4):2100138, 2022. doi: 10.1002/minf.202100138.

[12] Brendan D. McKay and Adolfo Piperno. Practical graph isomorphism, II. *Journal of Symbolic Computation*, 60: 94–112, 2014. doi: 10.1016/j.jsc.2013.09.003.

[13] Eric F. Pettersen, Thomas D. Goddard, Conrad C. Huang, et al. UCSF ChimeraX: Structure visualization for researchers, educators, and developers. *Protein Science*, 30:70–82, 2021. doi: 10.1002/pro.3943.

[14] S. A. Rahman, G. Torrance, L. Baldacci, S. Martínez Cuesta, F. Fenninger, N. Gopal, S. Choudhary, J. W. May, G. L. Holliday, C. Steinbeck, and J. M. Thornton. Reaction decoder tool (RDT): extracting features from chemical reactions. *Bioinformatics*, 32(13):2065–2066, 2016. doi: 10.1093/bioinformatics/btw096.

[15] Philippe Schwaller, Benjamin Hoover, Jean-Louis Reymond, Hendrik Strobelt, and Teodoro Laino. Extraction of organic chemistry grammar from unsupervised learning of chemical reactions. *Science Advances*, 7(15):eabe4166, 2021. doi: 10.1126/sciadv.abe4166.

A Evaluation details

A.1 Reproducibility and comparison scope

The repository's `manuscript/REPRODUCIBILITY.md` links the benchmark scripts, pinned software and input versions, full configurations, and per-case records. It distinguishes the completed Golden validation in Section 3.2 from the archived September evaluations below. Table 1 gives the archived comparison settings. The guide `docs/PYTHON_API.md` demonstrates matching, family reduction, separate local decoding and conditional symmetry queries through the Python API.

For the archived coordinate comparison, MAPPA used supplied WBO matrices and SLAP used the original XYZ-derived binary graphs. SLAP's returned heavy-atom assignments received hydrogen refinement under its original bond-change objective, then were rescored on the WBO matrices. Thus, the comparison uses a common event score but different input representations. It covers the returned SLAP witnesses, not an exhaustive expansion of its hydrogen label families. The WBO-generation method is not recorded in the available metadata, so regeneration of these matrices from molecular structures is not established.

The archived paired search timings use an Apple M4 Pro CPU. Averages use 1,821 common completed reactions. Reference verification and interrupted work are excluded. MAPPA includes checkpoint I/O; SLAP excludes I/O. Higher-seed runs measured on different CPUs are excluded from this paired comparison. For the archived coordinate stage comparison, MAPPA search includes checkpoint I/O; SLAP mapping excludes input preparation and export, and its hydrogen refinement includes I/O. Shared event rescoring and failed or replaced attempts are excluded. Archived Golden MAPPA timing includes symmetry finalization; SLAP includes graph construction, mapping, and export. In that primary Golden evaluation, uncut and swept SLAP have one and three unresolved cases, respectively; all MAPPA verdicts are resolved. These timing cohorts are not pooled with the current-design validation.

A.2 Archived default comparator protocol

Released comparators receive the same label-free Golden inputs, without prediction repair or reference-guided ranking. Invalid outputs and failures remain in the denominator. SLAP is evaluated over all returned candidates, using binary mode for the default comparison and both modes for bidirectional comparisons. The repository records the released models, versions, and adapter settings.

A.3 Archived event-window query scope

For each coordinate reaction, the archived event-window limit was one event above the larger of the native-SLAP and SLAP-sweep minima. All 120,052 retained families were exhausted within these limits, without a class-count cap. Higher-event patterns are outside this completeness claim. An event class records the signed atom-pair changes, modulo source permutations preserving every thresholded bond-change test against the opposite endpoint. Each class has a concrete mapping witness. This explicit whole-reaction query is separate from ordinary decoding into shared local fragment responses.

Table A1 Archived complete event-window queries on all 140 reactions. Every retained family is certified within the listed per-case event limit.

Maximum events	1	2	3	4	5	6	7	8	9
Reactions (all complete)	6	11	39	32	33	10	7	1	1

A.4 Preservation of mapping families under reduction and decoding

Family reduction retains each terminal representative, its required fragment constraints, and its allowed action sequence after applying prefix locks. Disjoint actions commute. Adjacent nested groups satisfy $HG = GH = G$ when $H \leq G$, with containment checked by exact generator membership. Other overlapping actions retain their order. Identical records are shared; branches with the same unordered fragment pairs retain the

union of their family records. The admissible mapping union is therefore unchanged. A shared fragment partition or automorphism group alone does not establish family equality.

Local decoding inventories all fragments before computing their responses. Sharing a response preserves the family constraints and any action that moves several fragments together. Each bond contributes once to a valid joint choice, and alternative partitions are alternative decompositions of the same family. This preserves the retained relation without requiring a whole-reaction event-class enumeration.

The archived event-window query separately certified a family through event invariance, a valid lower bound excluding the window, or unsatisfiability after excluding all discovered event classes. An exhausted budget remained incomplete. Lossless family representation, local event decoding, complete event-window queries and exhaustive structural search are distinct claims.

A.5 Archived Golden nonrecoveries

Table A2 The eleven archived ten-seed Golden nonrecoveries. Case indices are zero-based. Retained pair counts cover every terminal in both directions and every saved cut at ten seed orderings and cap 100. The four cardinality mismatches concern exact reference relations, including unmatched atoms. This is not the nonrecovery set of the current one-seed validation.

Case	Reference pairs	Retained pairs	Observed mismatch
7	29	30	One extra matched pair
19	23	23	Atom pairing excluded by orbit bounds
590	31	31	Atom pairing excluded by orbit bounds
871	20	20	Atom pairing excluded by orbit bounds
986	16	16	Atom pairing excluded by orbit bounds
1228	12	12	Atom pairing excluded by orbit bounds
1285	14	14	Joint C/O correspondence excluded
1358	36	37	One extra matched pair
1377	22	23	One extra matched pair
1475	18	19	One extra matched pair
1553	20	20	Atom pairing excluded by orbit bounds

Independent reconstruction agrees with the supplied references in all eleven cases. Four require a different number of matched pairs; seven require correspondences absent from the retained families. All have at least one capped direction, but this does not identify the cap as the cause. The uncapped traces discussed in the main text demonstrate greedy-growth exclusions in two cases. Reference-directed feasibility checks are diagnostic and do not contribute to benchmark recovery. Per-case certificates and traces are linked in the repository.

A.6 Archived branch-cap and CPU-time ablation on Golden

The archived study ran the twelve configurations in Table A3 on all 1,851 Golden reactions using an AMD EPYC 9J14 CPU. Both search directions are unioned. SLAP unions its binary and weighted modes. Search and reference verification have separate fixed resource budgets. The recorded cap-100 recovery counts reproduce the archived main evaluation. At cap 2,000, unresolved counts increase with seed count. Among reactions resolved at both caps, no swept configuration gains or loses a reference recovery.

The ten-seed cap-2,000 run leaves five references recovered at cap 100 unresolved, accounting for its lower confirmed count. The bounded SLAP sweep confirms 1,794 references here, compared with 1,796 in the main evaluation: two previously recovered cases are unresolved under this study's limits. These outcomes are reported separately from verified exclusions. The paired timing cohort excludes 448 reactions because at least one configuration does not finish; the all-attempt cost column exposes the additional computational cost on those difficult reactions. Source pins, individual outcomes, directional costs, and reproduction scripts are provided with the code.

Table A3 Archived Golden branch–cap ablation with paired SLAP baselines. Recovery counts confirmed reference inclusion over all 1,851 records; unknown cases remain in the denominator. CPU values are seconds per reaction. Paired CPU uses the same 1,403 reactions whose searches finish under every configuration. Spent CPU averages recorded search cost over all 1,851 selected attempts, including interrupted calls; it is not completed-search latency. Both timings exclude reference verification and initial imports. Superseded attempts are excluded. Cap 100 with a one-seed sweep is the default setting.

Method	Cap	Seeds	Recovered (%)	Unknown	Paired CPU	Spent CPU
MAPPA uncut	100	1	1,489 (80.44)	0	0.040	0.085
MAPPA sweep (default)	100	1	1,834 (99.08)	0	0.963	3.873
MAPPA sweep	100	2	1,834 (99.08)	0	1.619	7.260
MAPPA sweep	100	3	1,837 (99.24)	0	2.265	10.556
MAPPA sweep	100	10	1,840 (99.41)	0	6.993	33.211
MAPPA uncut	2,000	1	1,508 (81.47)	5	0.086	0.517
MAPPA sweep	2,000	1	1,834 (99.08)	2	4.317	30.055
MAPPA sweep	2,000	2	1,834 (99.08)	4	8.283	48.549
MAPPA sweep	2,000	3	1,837 (99.24)	6	12.201	61.890
MAPPA sweep	2,000	10	1,835 (99.14)	12	43.843	113.777
SLAP uncut	–	–	1,661 (89.74)	1	0.105	0.659
SLAP sweep	–	–	1,794 (96.92)	5	2.793	9.895

A.7 Archived end-to-end timing on coordinate reactions

The historical study measured one-seed, cap-2,000 cut-sweep search followed by its event-window decoder on the Apple M4 Pro CPU. CPU times are reported per reaction. The event thresholds and windows match the bond-event comparison, with no class-count cap. All 140 completed reactions reproduce the archived event-class sets and family counts. Median end-to-end wall time is 0.96 seconds. Timing includes input loading, checkpoint I/O, final-family construction, decoding, and candidate serialization; initial imports and reference-set verification are excluded. All 140 reactions finish within the five-minute watchdog. Detailed timing records and reproduction instructions are provided with the code.

Table A4 Archived MAPPA search plus event-window queries. CPU seconds per completed reaction in the 140-reaction pass. The final row measures the whole pipeline; medians and percentiles of individual stages are not additive. Completion coverage and timing scope are stated above.

Stage	Mean CPU s	Median CPU s	95th pct. CPU s
MAPPA sweep search	1.257	0.300	3.468
Final-family catalogue	0.156	0.019	0.595
Bond-event decoding	4.647	0.288	24.361
Candidate serialization	0.005	0.001	0.015
End to end	6.073	0.812	25.470